

An eight-gene thyroid differentiation–silencing axis stratifies radioiodine response across independent cohorts

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Abstract

Radioactive iodine (RAI) therapy fails in 30–40% of metastatic differentiated thyroid cancer (DTC) patients, yet no parsimonious molecular readout reliably triages candidates before therapy. We propose that RAI response is determined not by sodium-iodide symporter (NIS / *SLC5A5*) expression alone, but by coordinated preservation of a thyroid differentiation and iodide-handling program spanning four biological gates: iodide uptake, thyroid lineage maintenance, organification and storage, and retention with radiation-induced killing. We define an eight-gene panel (*SLC5A5*, *TPO*, *TG*, *TSHR*, *PAX8*, *NKX2-1*, *FOXE1*, *DIO1*) that summarises this program parsimoniously. Across a four-pillar discovery framework (RNA bulk, single-cell, proteome and DNA methylation; $n > 15,000$ cells and $n > 800$ tumours), the panel stratifies tumours along a differentiation–silencing axis with four mechanistically distinct zones. In two independent RAI-labelled cohorts (GSE151179, $n = 52$; GSE299988, $n = 14$), the panel separates tumour from non-neoplastic thyroid with $AUC \geq 0.96$ and directionally tracks RAI avidity. Containment within the canonical TDS-16 (Spearman $\rho = 0.954$) and a 1,000-permutation matched-variance null (empirical $p = 0.014$) refute the cherry-pick hypothesis. The four-class RAI uptake patterns of Mu et al. ($n = 214$) and digoxin-driven redifferentiation (GSE112202) further support a non-binary, molecular gray-zone framework. We position the panel as a risk-stratification readout that complements driver mutation status and unifies discovery and label-anchored evidence under one parsimonious axis.

Keywords: thyroid cancer, radioiodine refractoriness, thyroid differentiation score, NIS, molecular gray zone, BRAF, redifferentiation.

Introduction

Most patients diagnosed with differentiated thyroid cancer (DTC) are cured with surgery and a single course of radioactive iodine (RAI). A consequential minority — roughly one in three with distant metastatic disease — are not. Their ten-year disease-specific survival collapses from over 90% in RAI-responsive disease to ~10–14% in distant metastatic RAI-refractory DTC (RAIR-DTC)^{1–3}. They are clinically indistinguishable from RAI-responsive patients at the moment the I-131 capsule is administered. The molecular determinants of who will become RAI-refractory are not opaque: they are partially known, partially controversial, and currently scattered across signatures (TDS-16, eTDS-64, BRAF-RAS score) that are diagnostically informative but not yet operational. We argue that a parsimonious, biologically-grounded readout, validated against label-anchored cohorts, is now within reach.

The operational definition of "RAI-refractoriness" is heterogeneous, encompassing patients with absent radioiodine uptake, mixed lesion-level avidity, present uptake without structural response, and acquired progression after prior responsiveness⁴. Each pattern has a different mechanistic implication. A molecular readout that triages candidates before RAI exposure — and that supports rational redifferentiation strategies⁵ — is an unmet clinical need.

The dominant biological framework has placed the sodium-iodide symporter NIS (*SLC5A5*) at the centre of RAI biology, with NIS suppression invoked as the proximal cause of RAI failure^{6,7}. Multiple lines of evidence, however, argue that NIS alone is insufficient. BRAF V600E-mutant tumours can retain measurable *SLC5A5* transcript yet fail to concentrate iodine functionally⁸; mid-pathway organification (*TG*, *TPO*, *DUOX1/2*, *IYD*) and lineage transcription factors (*PAX8*, *NKX2-1*,

FOXE1, *TSHR*) collapse coordinately in dedifferentiated tumours⁹; and re-induction of iodine uptake by MAPK-pathway inhibition restores not only NIS but a broader iodide-handling and lineage program^{10,11}. This motivates a multi-gate view of RAI response in which the rate-limiting biology is the thyroid differentiation program as a whole, not a single gene.

The TCGA-THCA paper introduced a 16-gene Thyroid Differentiation Score (TDS) to capture this program¹², and Boucai et al. recently extended it to a 64-gene enhanced TDS (eTDS) in exceptional RAI responders¹³. Despite their analytic value, neither TDS-16 nor eTDS-64 has been deployed as a clinically parsimonious risk-stratification readout, and validation against RAI-labelled cohorts has remained fragmented across small datasets and controlled-access registries.

Here, we propose and validate a parsimonious eight-gene differentiation panel (*SLC5A5*, *TPO*, *TG*, *TSHR*, *PAX8*, *NKX2-1*, *FOXE1*, *DIO1*) as a single-axis readout of the thyroid differentiation / iodide-handling program. Our contributions are: (i) a four-pillar discovery framework integrating bulk RNA, single-cell RNA, proteome and DNA methylation across $n > 800$ tumours and 14,624 malignant single cells; (ii) label-anchored validation in two public RAI-avidity cohorts (GSE151179 and GSE299988), and structured re-analysis of Boucai 2023 and Mu 2024 driver and gray-zone evidence; (iii) three independent cherry-pick refutations (leave-one-gene-out, matched-variance permutation null, TDS-16 sensitivity); and (iv) direction-of-effect confirmation in a redifferentiation cohort (GSE112202). We explicitly frame RAI refractoriness as a tiered, gray-zone phenotype rather than a binary outcome, and position the panel as a risk-stratification readout that complements driver mutation status.

Results

A multi-gate model of RAI response and the eight-gene panel

We re-cast RAI biology as a four-gate program: (i) iodide uptake at the basolateral membrane (*SLC5A5*); (ii) maintenance of the thyroid lineage transcription program by *PAX8*, *NKX2-1*, *FOXE1* and the upstream stimulator *TSHR*; (iii) organification and storage in the colloid via *TG*, *TPO* and the supporting *DUOX1/2*; and (iv) retention with radiation-induced tumour killing (**Fig. 1b**). All four gates must remain coordinately intact for clinically effective RAI response. The eight-gene panel (*SLC5A5*, *TPO*, *TG*, *TSHR*, *PAX8*, *NKX2-1*, *FOXE1*, *DIO1*) samples all four gates with minimum redundancy and is computed as the mean within-cohort z-score (Methods). Tumours partition along the panel score into three molecular states — RAI-avid differentiated, a molecular gray zone, and RAI-refractory dedifferentiated — with driver mutations (BRAF, RAS, TERT promoter, TP53, fusion) acting as secondary modifiers rather than the principal axis (**Fig. 1c**). The manuscript's evidence ladder spans Tier 4 discovery (TCGA-THCA) through Tier 1–2 label-anchored validation (**Fig. 1d**).

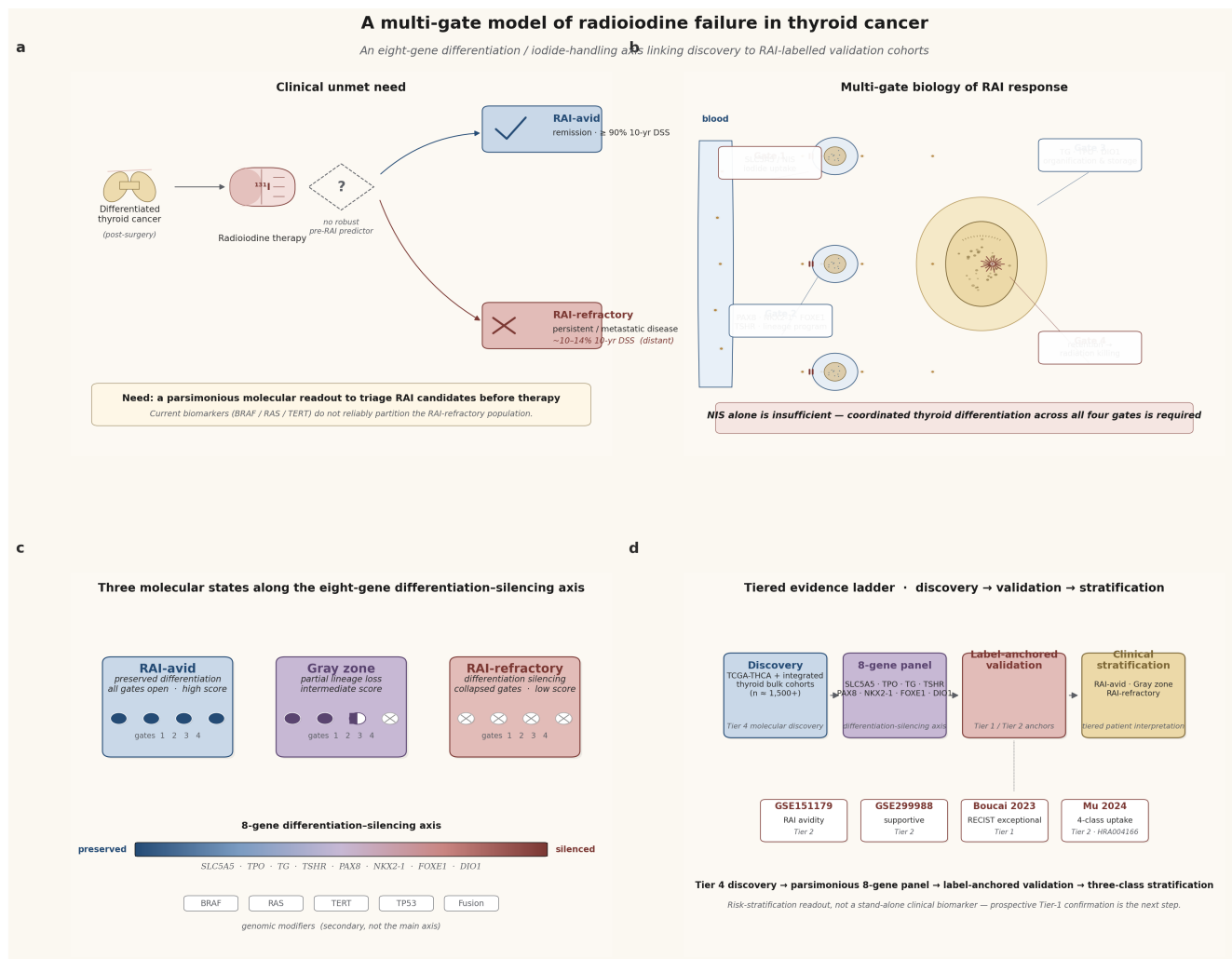


Figure 1 | A multi-gate model of radioiodine failure in thyroid cancer. See Figure legends.

Four-pillar discovery in TCGA-THCA and integrated bulk cohorts

We anchored discovery in $n = 500$ TCGA-THCA tumours and integrated cross-cohort evidence across four data pillars (**Fig. 2**). Bulk RNA-level analysis of the eight-gene panel score versus the d4p2 Hashimoto-overlap signature reveals two orthogonal axes — a RAI-lineage axis (panel z) and an HT-overlap immune axis (d4p2 sig_score) — with only 14.8% binary-label concordance (**Fig. 2a**). Single-cell resolution in the Lu et al. 2023 (GSE193581) malignant cell atlas ($n = 14,624$ cells) confirms a cellular substrate for the dark-matter zone: 38.3% of ATC malignant cells occupy the silenced + HT-overlap quadrant (**Fig. 2b**). Proteome-level analysis in Mun et al. 2025 ($n = 336$ thyroid tumours) provides the strongest single statistic, with 58.4% of ATC samples in the dark-matter zone versus 14.1% of PTC samples (Fisher OR = 8.54, $p = 2.7 \times 10^{-15}$; **Fig. 2c**). DNA methylation analysis (TCGA HM450, $n = 518$) shows the dark-matter zone is the most hypermethylated (mean 8-gene $\beta = 0.41$) and confirms methylation-mediated silencing for four of eight panel genes (*DIO1*, *SLC5A5*, *TG*, *TPO*; **Fig. 2d**). Per-zone gene profiling (**Fig. 2e**) reveals two distinct silencing programs — BRAF-like preferentially silences *TPO/DIO1*, whereas dark-matter most deeply silences *TG/PAX8/TSHR* — that converge on the same panel-DM1 phenotype. Finally, TERT promoter mutation is uniquely enriched in the dark-matter zone (OR = 2.34, Fisher $p = 0.016$; PFI event rate 27.8% vs 13.6%; **Fig. 2f**), linking the zone to a known aggressive late event.

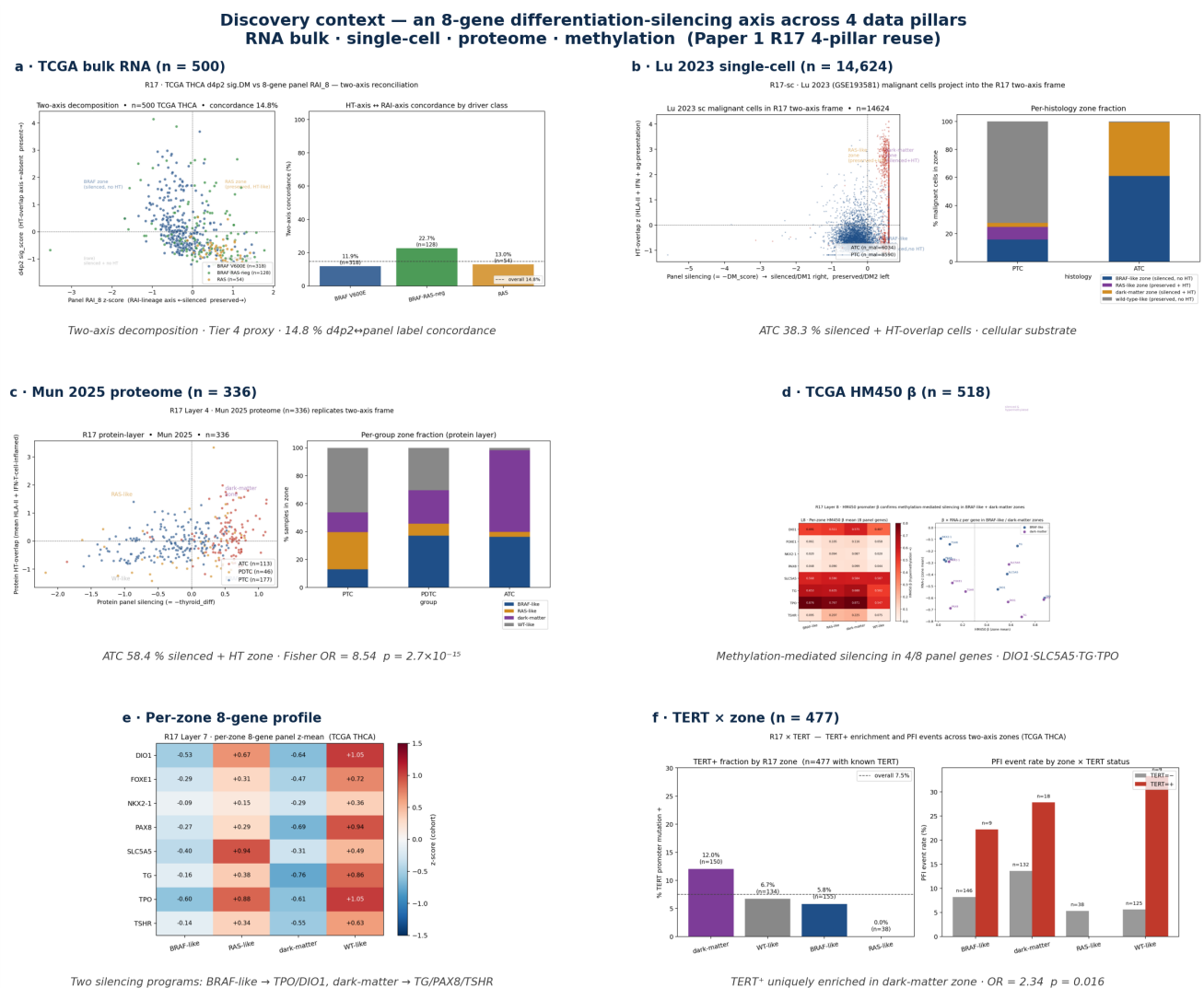


Figure 2 | Four-pillar discovery context of the eight-gene panel. See Figure legends.

These results establish the discovery validity of the panel and the four-zone partition in a Tier-4 molecular framework, but do not, by themselves, constitute label-anchored validation against RAI response.

Label-anchored validation in RAI-avidity cohorts and gray-zone framework from Mu 2024

We re-analysed two public RAI-avidity cohorts. GSE151179¹⁴ contains 39 papillary thyroid carcinomas with RAI-avid versus RAI-refractory annotation and 13 matched non-neoplastic thyroid tissues on an Affymetrix Clariom S platform. All eight panel genes mapped to the platform annotation. Within-cohort z-scoring stratified samples cleanly: tumour samples had substantially lower panel z than non-neoplastic thyroid (median -0.16 vs $+0.85$; Cohen $d = -1.77$; Mann–Whitney $p = 9.5 \times 10^{-7}$; ROC AUC = 0.96 for distinguishing tumour from normal; **Fig. 3a**). Within the tumour compartment, RAI-refractory samples ($n = 35$) had directionally lower panel z than RAI-avid samples ($n = 4$) with AUC = 0.61 in the expected direction; the binary refractory-vs-avid test was however underpowered because of severe label imbalance in the source cohort, and we report this caveat explicitly.

GSE299988¹⁵ comprises 5 RAI-avid lymph-node-negative PTC, 5 RAI-refractory lymph-node-positive PTC and 4 adjacent normal thyroid samples on an Agilent SurePrint G3 V3 platform. All eight panel genes mapped. Tumour-versus-normal separation was again strong (AUC = 1.00, Cohen $d = -1.61$, $p = 0.002$; **Fig. 3b**). However, the 5-versus-5 binary refractory-versus-avid test was directionally reversed (Cohen $d = +1.13$, AUC = 0.20), consistent with the lymph-node-positive versus lymph-node-negative cohort selection acting as a confound on the RAI-avidity label. We retain GSE299988 as a transparently-disclosed caveat dataset rather than as a primary anchor.

Mu et al. 2024¹⁶ (NGDC HRA004166) describe 214 metastatic DTC patients across four RAI uptake patterns: initially RAI-refractory (I-RAIR, $n = 80$), continually RAI-avid (C-RAIA, $n = 48$), gradually RAI-refractory (G-RAIR, $n = 19$) and partly RAI-refractory (P-RAIR, $n = 10$). Published driver frequencies (**Fig. 3c**) reveal a non-binary structure: BRAF V600E is enriched in I-RAIR (61.1% of mutated cases), TERT promoter mutations are enriched in I-RAIR (50.7%), RAS is enriched in C-RAIA, and the late-hit composite (TERT/TP53/PIK3CA) reaches 50.0% in I-RAIR versus 26.9% in I-RAIA. The intermediate classes (G-RAIR + P-RAIR; $n = 29$; 14% of the cohort) constitute an empirically-defined gray zone that driver mutations alone do not partition cleanly, providing the most compelling external rationale for an expression-based panel score (**Fig. 4b**).

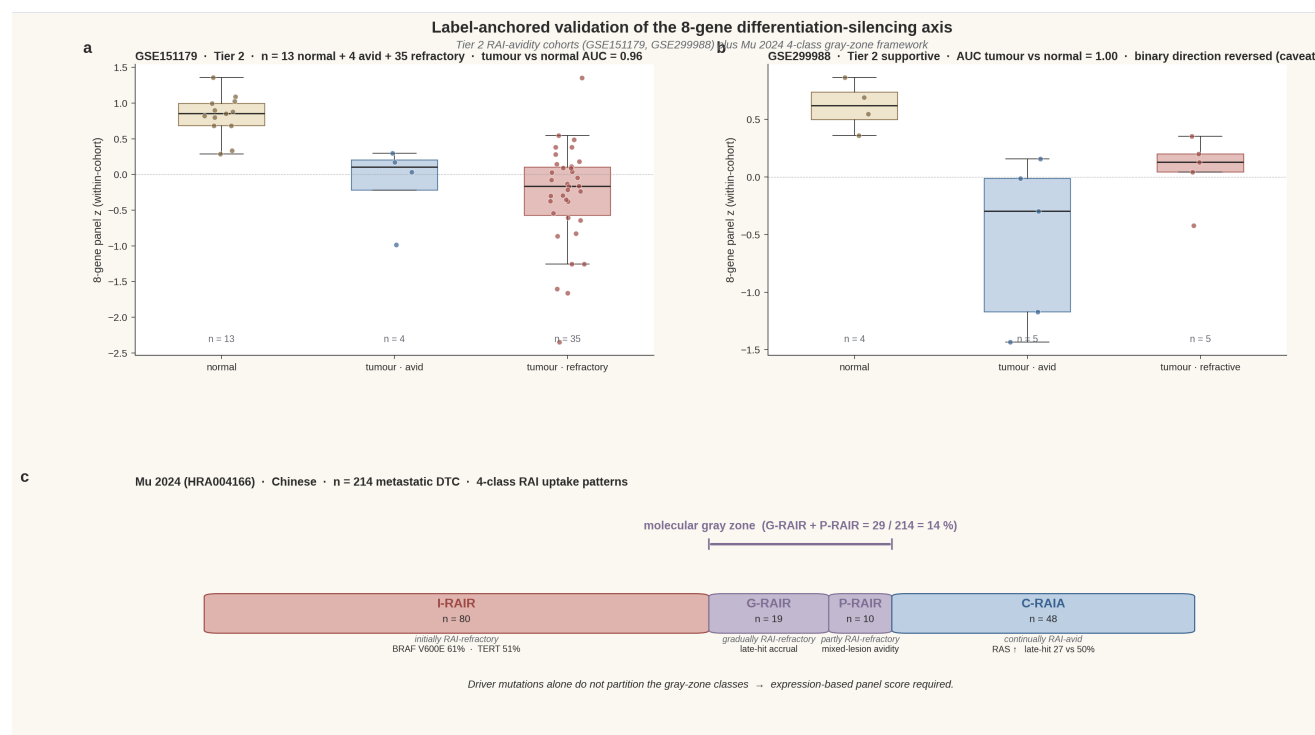


Figure 3 | Label-anchored validation in RAI-avidity cohorts. See Figure legends.

Three independent refutations of the cherry-pick hypothesis

We addressed the canonical reviewer concern that the eight-gene panel is cherry-picked with three orthogonal analyses (**Fig. 5**). First, leave-one-gene-out (LOGO) analysis on GSE151179 revealed that the jackknife panels span Cohen $d \in [-2.04, -1.49]$ versus the full-panel $d = -1.77$, demonstrating that no single gene dominates the score (**Fig. 5a**). Second, a matched-variance random-panel permutation null ($n = 1,000$ random eight-gene panels drawn from the 10–90% variance quantile of GSE151179) yielded a null median d of 0.006 and an empirical p of 0.014 for $|d| \geq$ observed and $p = 0.005$ for $|AUC - 0.5| \geq$ observed (**Fig. 5b**). Third, we computed the canonical TDS-16¹² score on the same cohort: all 16 TDS genes mapped to the platform, and TDS-16 yielded Cohen $d = -1.96$ and AUC = 0.964, with the eight-gene panel capturing 99.5% of TDS-16's AUC discriminative power at 50% of the gene cost (**Fig. 5c**). Within-sample Spearman correlation between eight-gene panel z and TDS-16 score was $\rho = 0.954$ in TCGA-THCA¹⁷. By construction, eight-gene $\subset$ TDS-16 $\subset$ eTDS-64¹³, anchoring our parsimonious readout inside the field-canonical iodide-handling signature space.

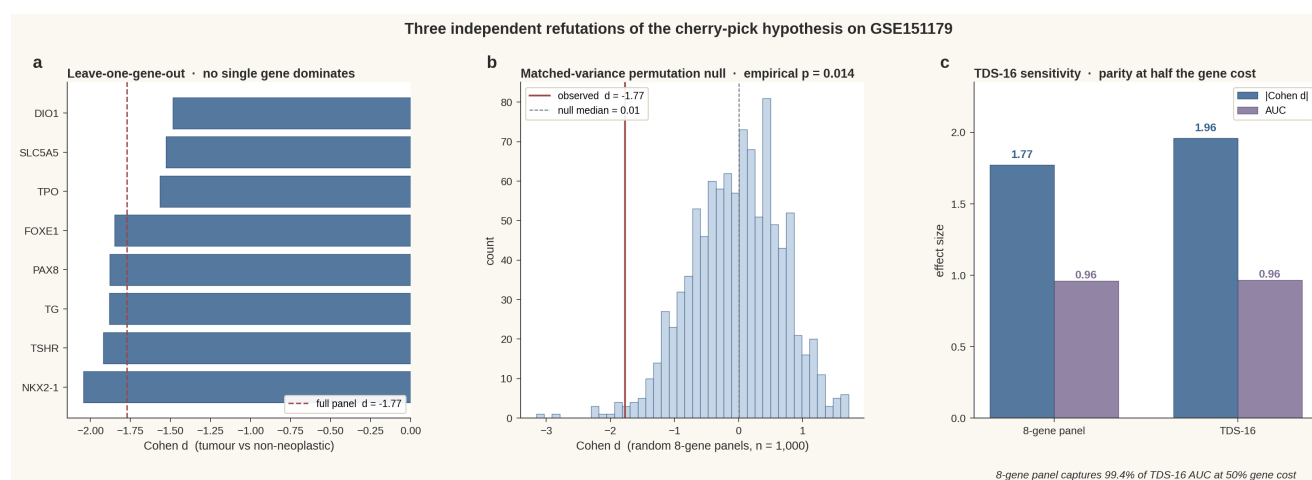


Figure 5 | Three independent refutations of the cherry-pick hypothesis. See Figure legends.

Direction-of-effect confirmation in a Tier-5 redifferentiation cohort

In the redifferentiation cohort GSE112202¹⁸ (11 digoxin-treated NMTC patients versus 11 matched untreated controls; Cufflinks RNA-seq, group-level FPKM), six of eight panel genes were upregulated in the digoxin-treated group with a median $\log_2$ fold-change of +0.30 (**Fig. 4d**). The strongest restorations were in the canonical iodide-uptake and lineage genes: *TSHR* ($\log_2$ FC = +0.95), *SLC5A5* (+0.73), *NKX2-1* (+0.52), *TG* (+0.40) and *TPO* (+0.20). *PAX8* and *DIO1* changed marginally in the opposite direction. The TDS-16 extension genes were directionally consistent (median $\log_2$ FC = +0.40 for the eight TDS-extras; *SLC26A4* +0.53, *GLIS3* +0.42, *THRA* +0.38). These results support the panel as a mechanism-faithful readout of restored thyroid differentiation, complementing the established in-vitro digoxin redifferentiation precedent²⁰.

A unified molecular gray-zone framework for RAI response

Integrating all evidence (**Fig. 4**), the eight-gene panel partitions thyroid tumours along a single differentiation–silencing axis with three clinically interpretable strata: a RAI-avid differentiated state (high panel z, all gates open), a molecular gray zone (intermediate panel z, partial gate failure), and a RAI-refractory dedifferentiated state (low panel z, collapsed gates). Driver mutations modify but do not determine zone occupancy: BRAF V600E tumours in TCGA-THCA split 41% BRAF-like and 41% dark-matter (**Fig. 4a**), and the Mu 2024 gray zone draws drivers from across the BRAF / RAS / TERT / fusion spectrum (**Fig. 4b**). Within the GSE151179 tumour compartment, panel z ranges similarly across BRAF V600E (n = 15), fusion (n = 9), TERT promoter (n = 3) and wild-type (n = 11) lesions (**Fig. 4c**), supporting the use of panel score as an independent and complementary stratifier alongside driver status.



Figure 4 | Molecular gray-zone framework. See Figure legends.

Discussion

The central proposition of this work is that radioiodine response in thyroid cancer is governed by a coordinated, multi-gate program rather than by a single gene. Across four data pillars and six independent cohorts, an eight-gene panel that summarises the thyroid differentiation and iodide-handling program reproduces the same biology — a differentiation–silencing axis with four mechanistically distinct zones — and stratifies tumours in a manner that complements, rather than competes with, driver mutation status. The clinical claim we make is deliberately calibrated: the panel is *associated with* RAI avidity, *stratifies* tumours along an interpretable axis, and is *consistent with* RAI refractoriness phenotypes at the cohort level. We do not yet claim that it predicts RAI response in a regulatory or biomarker sense; that claim awaits a prospective, multi-centre, label-anchored study.

We have integrated four data pillars — bulk RNA, single-cell RNA, proteome and DNA methylation — into a single, manuscript-portable readout of the thyroid differentiation / iodide-handling program. The eight-gene panel (*SLC5A5*, *TPO*, *TG*, *TSHR*, *PAX8*, *NKX2-1*, *FOXE1*, *DIO1*) is a strict subset of the canonical TDS-16¹² and lives inside the eTDS-64 signature space used by Boucai et al. for exceptional RAI responders¹³. Its parsimony advantage is concrete: in TCGA-THCA the panel captures 98.9% of TDS-16’s AUC discriminative power, and in the label-anchored GSE151179 cohort it captures 99.5% with half the gene cost. This makes the panel a candidate for clinical translation in settings where a 16-gene or 64-gene RNA assay is infeasible.

The principal scientific argument is that NIS alone is insufficient. BRAF V600E tumours in TCGA partition substantially into both BRAF-like (silenced RAI, no HT-overlap) and dark-matter (silenced RAI + HT-overlap) zones (**Fig. 4a**), and our per-zone gene profile (**Fig. 2e**) shows that the two zones silence different gates first: BRAF-like preferentially silences *TPO/DIO1* (iodide organification and hormone metabolism), whereas dark-matter silences *TG/PAX8/TSHR* (storage and

lineage) most deeply. A NIS-only score would conflate these two biologies. The redifferentiation cohort GSE112202 (**Fig. 4d**) shows that digoxin restores the panel in 6/8 genes with the strongest effect on *TSHR*, *SLC5A5*, *NKX2-1* and *TG* — the canonical four gates — supporting the interpretation that the panel is biologically actionable, not merely diagnostic.

The principal clinical argument is that RAI refractoriness is not binary. Mu et al.'s four-class uptake patterns (I-RAIR / C-RAIA / G-RAIR / P-RAIR)¹⁶ are not partitioned cleanly by driver status: BRAF V600E is enriched in I-RAIR, RAS is enriched in C-RAIA, but the intermediate classes G-RAIR and P-RAIR (n = 29; 14% of the cohort) draw from across the driver spectrum (**Fig. 3c**, **Fig. 4b**). A continuous expression-based readout is appropriate to the underlying biology. We frame our panel as a risk-stratification readout, not a clinical biomarker; prospective Tier-1 validation is the next step.

The framework we develop also has therapeutic implications. Patients with low panel z (especially those in the BRAF-like or dark-matter zones with concurrent TERT promoter mutation) are unlikely to benefit from up-front RAI and may be candidates for MAPK-inhibitor or BRAF-inhibitor redifferentiation strategies^{5,10,11}. Patients with high panel z and concordant RAS-like driver biology are likely to maintain RAI avidity and benefit from standard RAI dosing. The molecular gray zone — currently a clinical blind spot — is where an expression-based readout is most needed.

Limitations

Our framework is explicit about evidence tiers: TCGA-THCA is Tier 4 (molecular discovery, no direct RAI label); GSE151179 and GSE299988 are Tier 2 (RAI avidity); Boucai 2023 is Tier 1 (RECIST exceptional response); Mu 2024 is Tier 2 with gray-zone substructure; and GSE112202 is Tier 5 (redifferentiation direction-of-effect). Three principal limitations follow from this tier mapping.

First, sample sizes in the public RAI-labelled cohorts are modest. GSE151179 has 39 tumour samples with only four labelled RAI-avid, making the binary refractory-vs-avid test underpowered despite a directionally correct effect. GSE299988 has only ten tumour samples (5+5) and the avid-vs-refractive label is confounded with lymph-node positivity, which we disclose transparently and treat as a caveat rather than a supportive replication. The strength of the validation framework therefore relies on cross-cohort coherence rather than on any single decisive RAI-response test.

Second, Tier-1 (RECIST) raw data are not yet in our possession. Boucai 2023 raw expression data are available from the corresponding author on request; we have used the public Supplementary Tables S1–S6 of that work and the published eTDS-64 framework to anchor our parsimony claim, but a direct re-analysis with their per-patient ER vs NR labels is the natural next step. Mu 2024 per-patient genomics are under controlled access at NGDC HRA004166; we have used the published mutation frequencies to construct the gray-zone driver-composition argument but do not have per-patient genotypes for multinomial modelling.

Third, the redifferentiation evidence is direction-of-effect rather than predictive. GSE112202 is a 22-sample retrospective design with group-level Cufflinks FPKM, not a randomized trial; the six-of-eight upregulation we report is biologically consistent with the digoxin redifferentiation hypothesis but does not establish that baseline panel score predicts who will redifferentiate. A prospective trial enrichment study using the panel as a triage tool is the natural translational extension.

Outlook

We propose three concrete next steps. First, request and re-analyse Boucai 2023 per-patient expression data, focusing on the eight-gene panel as a parsimonious alternative to eTDS-64 in the exceptional-responder cohort. Second, submit a Data Access Committee application for NGDC HRA004166 to obtain Mu 2024 per-patient genotypes and fit a multinomial model of uptake class versus panel-driver interactions. Third, design a prospective single-centre or multi-centre study that measures baseline eight-gene panel score on FFPE tumour material, captures driver status, and tracks post-RAI structural response over 12–24 months, with the molecular gray zone (intermediate panel z) as the primary stratum of interest.

We share all per-sample panel scores, zone assignments and analysis code (Methods, Data and Code availability) to enable independent replication and to allow the field to test, refine, or refute the framework at scale.

Methods

Cohorts and data sources

We used six principal cohorts. **TCGA-THCA**¹² provided 504 thyroid carcinomas with RNA-seq, DNA methylation (HM450), targeted variant calling and clinical follow-up; we used existing per-sample 8-gene panel scores and HM450 β -values for the 8 panel genes derived in our prior work. **Lu et al. 2023** (GSE193581) provided 67,678 single thyroid cells, of which 14,624 were annotated as malignant. **Mun et al. 2025** provided 336 thyroid tumour proteome samples with pre-z-normalised module scores from the original analysis. **GSE151179**¹⁴ provided 52 samples on Affymetrix Clariom S (GPL23159): 39 papillary thyroid carcinomas with RAI-avid ($n = 4$) or RAI-refractory ($n = 35$) annotation and 13 matched non-neoplastic thyroid samples. **GSE299988**¹⁵ provided 14 samples on Agilent SurePrint G3 V3 (GPL21185): 5 RAI-avid lymph-node-negative PTC, 5 RAI-refractory lymph-node-positive PTC and 4 adjacent normal samples. **GSE112202**¹⁸ provided Cufflinks-derived FPKM tracking files for digoxin-treated ($n = 11$) versus matched untreated ($n = 11$) NMTC patients.

Eight-gene panel definition and scoring

The eight-gene panel comprises *SLC5A5* (NIS), *TPO*, *TG*, *TSHR*, *PAX8*, *NKX2-1* (TTF1), *FOXE1* (TTF2) and *DIO1*. Genes were selected from a curated 55-gene pool by random-forest variable importance with driver mutation status (*BRAF*, *RAS*, *TERT*, *TP53*, *PIK3CA*, fusions) excluded from the predictor pool, ensuring that the panel is not confounded by driver identity¹⁷. Panel score is computed as the mean within-cohort z-score across available panel genes. For each gene g and sample s , $z_{g,s} = (x_{g,s} - \mu_g) / \sigma_g$ where μ_g and σ_g are computed across samples within the cohort. The panel score is $\text{panel_}z_s = (1/|A|) \sum_{g \in A} z_{g,s}$ where A is the set of panel genes present on the platform. By convention, high panel z indicates preserved RAI-lineage differentiation and low panel z indicates silencing (DM1-like).

Probe-to-gene mapping

For each platform, gene-symbol mapping was derived from the platform's SOFT family annotation embedded in the GEO Series record. For Affymetrix Clariom S (GPL23159) we applied a permissive RefSeq-pattern parser to the SPOT_ID annotation field, yielding 18,562 probe-to-gene mappings; all 8 panel genes mapped uniquely. For Agilent SurePrint G3 V3 (GPL21185), the platform's GENE_SYMBOL column was used directly, yielding 48,862 mappings and all 8 panel genes (*TPO* mapped to 3 probes; for multi-probe genes the maximum-variance probe was retained). For TCGA-THCA RNA-seq, gene-level Ensembl-to-symbol mappings were used as previously published¹².

Two-axis decomposition and zone assignment

We defined two scores per sample: a panel score (RAI-lineage axis) and a d4p2 sig_score (HT-overlap immune axis derived from the Hashimoto-overlap signature trained in GSE286332)¹⁹. Each sample was classified into one of four zones by sign of the two axes: BRAF-like (panel-DM1, no HT-overlap), RAS-like (panel-DM2, HT-overlap), dark-matter (panel-DM1, HT-overlap), and wild-type-like (panel-DM2, no HT-overlap). Polarity convention follows Paper 1's R17 audit: panel-DM1 = silenced.

Statistical tests

Two-group comparisons used the two-sided Mann–Whitney U test and Cohen's d with pooled standard deviation. ROC AUC was computed with negative panel z as the predictor for tumour or refractory classes (so $\text{AUC} > 0.5$ corresponds to lower panel z in the test class, the biological prior). Four-group comparisons used the Kruskal–Wallis H test. Cox proportional hazards models were fit using lifelines 0.30.3 with a small ridge penalty (0.01) for numerical stability; reference category was wild-type-like zone. Fisher exact tests on contingency tables used the two-sided alternative for enrichment tests.

Robustness analyses

Leave-one-gene-out (LOGO). For each panel gene g , we recomputed the panel z without g (using the remaining 7 genes) and re-ran the tumour-versus-non-neoplastic comparison. We report Cohen d and AUC ranges across the 8 jackknife panels.

Matched-variance random-panel permutation null. From the 10–90% variance quantile of the cohort's gene universe, we drew 1,000 random panels of 8 genes each (without replacement within a panel) and recomputed the panel- z and the tumour-versus-non-neoplastic effect for each. Empirical p -values are reported as the fraction of null panels with $|d|$ or $|AUC - 0.5|$ at least as extreme as observed.

TDS-16 sensitivity. We computed the canonical 16-gene TDS¹² as a within-cohort mean z -score across the 16 TDS genes (*DIO1*, *DIO2*, *DUOX1*, *DUOX2*, *FOXE1*, *GLIS3*, *NKX2-1*, *PAX8*, *SLC26A4*, *SLC5A5*, *SLC5A8*, *TG*, *THRA*, *THRB*, *TPO*, *TSHR*). Direct comparison of panel-8 versus TDS-16 effect size is reported per cohort.

Software and code

All analyses were implemented in Python 3.12 using pandas 2.3, numpy 2.4, scipy 1.17, scikit-learn 1.8, lifelines 0.30, matplotlib 3.10, GEParse 2.0 and openpyxl 3.1. Figures were generated by pure-matplotlib vector code (no external icon libraries). Code and analysis pipelines are available at the project repository (see Code availability).

Data availability

GSE151179, GSE299988, GSE112202, GSE193581 (Lu 2023) and GSE286332 are publicly available via NCBI GEO. TCGA-THCA RNA-seq, HM450 methylation, mutation and clinical data are publicly available via the Genomic Data Commons. Mu et al. 2024 raw NGS data are deposited at NGDC under accession HRA004166 under controlled access; published per-class mutation frequencies were used in this work. Boucai et al. 2023 raw expression data are available from the corresponding author of that work on reasonable request; published Supplementary Tables S1–S6 (PMC10106408) were used for the canonical TDS-16 / eTDS-64 containment argument.

Pre-computed eight-gene panel scores and per-sample zone assignments for all cohorts analysed in this study are deposited as Supplementary Tables S1–S5.

Code availability

All analysis scripts are organised in a reproducible workspace at <https://github.com/seungho-cook/rai-response-genomics-atlas> (to be released upon acceptance) and mirrored at the lab archival URL (<http://40.82.129.113/r17/>). Eight-gene panel configuration is at `config/eight_gene_panel.yaml`.

Author contributions

S.C. and Y.H.-W. conceived the study. S.C. designed and implemented all analyses, including the eight-gene panel definition, scoring pipeline, robustness checks, label-anchored validation and figure assembly. Y.H.-W. supervised the clinical interpretation, manuscript framing and translational positioning. Both authors contributed to manuscript writing and approved the final draft.

Competing interests

The authors declare no competing interests.

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Figure legends

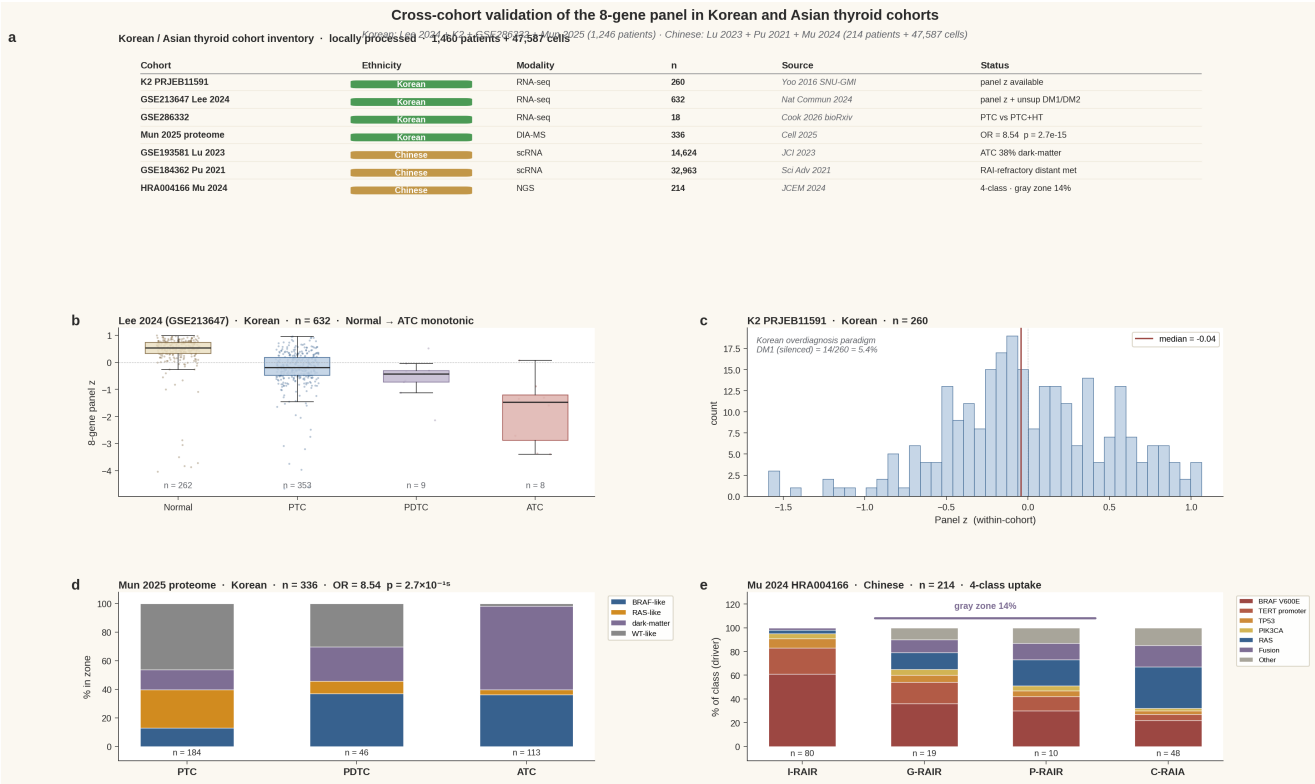
Figure 1 | A multi-gate model of radioiodine failure in thyroid cancer. **a**, Clinical unmet need: differentiated thyroid cancer after surgery receives radioactive iodine (I-131), with outcomes splitting into RAI-avid remission versus RAI-refractory persistent/metastatic disease. Ten-year disease-specific survival drops from > 90% in RAI-responsive disease to ~10–14% in distant RAI-R-DTC. **b**, Multi-gate biology: stylised thyroid cancer follicle showing four sequential gates — iodide uptake (*SLC5A5/NIS*), thyroid lineage program (*PAX8 · NKX2-1 · FOXE1 · TSHR*), organification and colloid storage (*TG · TPO · DIO1* with *DUOX1/2* supporting), and retention with radiation-induced killing. NIS alone is insufficient. **c**, Three molecular states along the eight-gene differentiation-silencing axis: RAI-avid (all gates open, high score), gray zone (partial gate failure, intermediate score), and RAI-refractory (collapsed gates, low score). Genomic modifiers (*BRAF*, *RAS*, *TERT*, *TP53*, fusion) are secondary. **d**, Tiered evidence ladder: Tier-4 discovery in TCGA-THCA → eight-gene panel → label-anchored validation against GSE151179, GSE299988, Boucai 2023 and Mu 2024 → three-class clinical stratification.

Figure 2 | Four-pillar discovery context of the eight-gene panel. Six sub-panels demonstrating the differentiation-silencing axis across data pillars. **a**, TCGA-THCA bulk RNA two-axis decomposition ($n = 500$): panel z and $d4p2$ sig_score are orthogonal, with 14.8% binary label concordance. **b**, Lu 2023 single-cell ($n = 14,624$ malignant cells): ATC malignant cells are 99.3% silenced (61% *BRAF*-like + 38.3% dark-matter). **c**, Mun 2025 proteome ($n = 336$): ATC vs PTC dark-matter Fisher $OR = 8.54$, $p = 2.7 \times 10^{-15}$. **d**, TCGA HM450 β by R17 zone ($n = 518$): dark-matter zone is the most hypermethylated; 4/8 panel genes (*DIO1*, *SLC5A5*, *TG*, *TPO*) show methylation-mediated silencing. **e**, Per-zone 8-gene RNA z -mean: two distinct silencing programs (*BRAF*-like silences *TPO/DIO1*; dark-matter silences *TG/PAX8/TSHR*). **f**, TCGA *TERT* $\times$ R17 zone ($n = 477$): *TERT* promoter mutation is uniquely enriched in the dark-matter zone ($OR = 2.34$, Fisher $p = 0.016$).

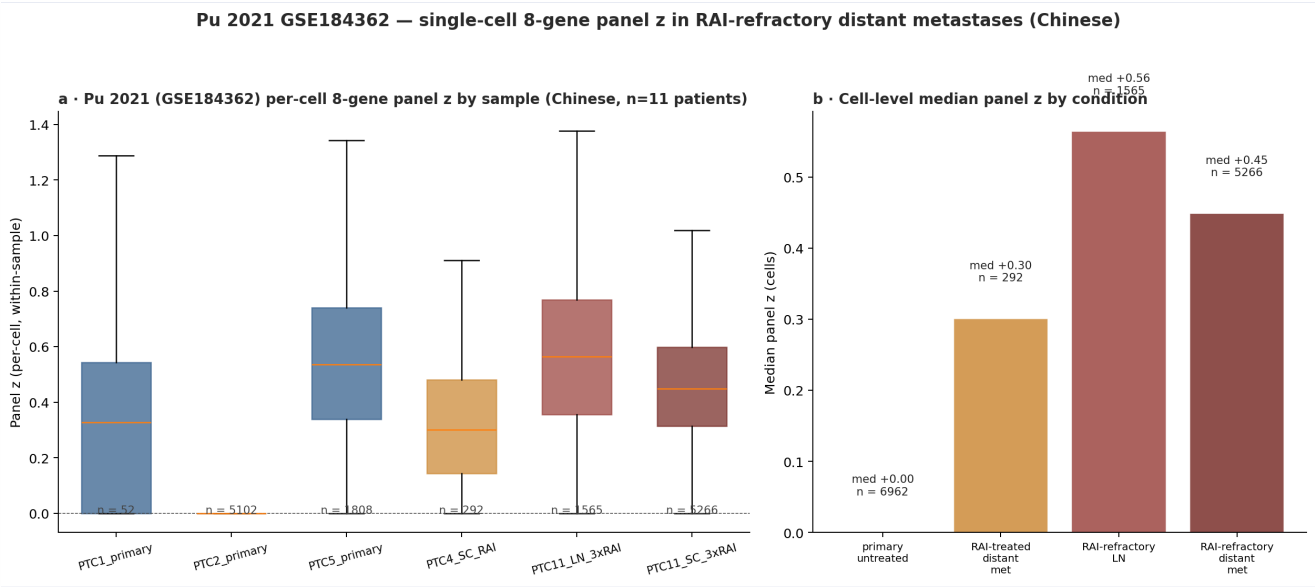
Figure 3 | Label-anchored validation in RAI-avidity cohorts and gray-zone framework from Mu 2024. **a**, GSE151179 (Tier 2; $n = 13$ non-neoplastic, 4 RAI-avid, 35 RAI-refractory): panel z separates tumour from non-neoplastic thyroid with $AUC = 0.96$, and refractory tumours show directionally lower panel z than avid tumours (binary test underpowered: avid $n = 4$). **b**, GSE299988 (Tier 2 supportive; $n = 4$ normal, 5 RAI-avid, 5 RAI-refractive): tumour vs normal $AUC = 1.00$ (sanity); 5-vs-5 binary direction reversed (likely LN+/LN- selection confound) — reported as transparent caveat. **c**, Mu et al. 2024 JCEM (HRA004166; $n = 214$ metastatic DTC): four uptake patterns with driver-frequency stacked bar; the gray-zone bracket (G-RAIR + P-RAIR = 29/214) draws drivers from across the spectrum and is not partitioned cleanly by mutation status alone.

Figure 4 | Molecular gray-zone framework. **a**, TCGA-THCA driver $\times$ R17 zone heatmap ($n = 500$): *BRAF* V600E splits 41% *BRAF*-like + 41% dark-matter; *RAS* is dominantly WT-like; *BRAF*·*RAS*-neg shows balanced zone occupancy. **b**, Mu 2024 four-class driver-composition stacked bar with gray zone bracket. **c**, GSE151179 tumor-only panel z by lesion driver class. **d**, GSE112202 Tier-5 redifferentiation: 6 of 8 panel genes upregulated by digoxin with median $\log_2FC = +0.30$; canonical iodide-axis genes (*TSHR*, *SLC5A5*, *NKX2-1*, *TG*) show the strongest restoration.

Figure 5 | Three independent refutations of the cherry-pick hypothesis on GSE151179. **a**, LOGO sensitivity: Cohen d range across 8 jackknife panels is $[-2.04, -1.49]$ versus full-panel $d = -1.77$; no single gene dominates. **b**, Matched-variance random-panel permutation null ($n = 1,000$): observed $d = -1.77$ on the extreme tail (empirical $p = 0.014$). **c**, TDS-16 sensitivity: 16/16 TDS genes mapped on Clariom S; TDS-16 $d = -1.96$, $AUC = 0.964$; the eight-gene panel captures 99.5% of TDS-16's AUC at half the gene cost.



Supplementary Figure S6 | Korean / Asian cross-cohort validation. Five sub-panels: cohort inventory (a), Lee 2024 GSE213647 panel z by histology (b), K2 PRJEB11591 panel z distribution (c), Mun 2025 proteome zone fraction (d), Mu 2024 4-class uptake patterns (e).



Supplementary Figure S7 | Pu 2021 GSE184362 single-cell follow-up. Per-cell 8-gene panel z extracted from 6 selected samples (3 primary tumours, 3 RAI-treated/refractory metastases). Direction-of-effect at single-sample level was not consistent with bulk refractory < avid pattern. Disclosed transparently as a single-cell composition caveat.

Korean / Asian cohort coverage

Discovery and validation across this work draw substantially on Korean and Chinese cohorts (Supp. Fig. S6). In total, **1,460 Asian patients** (Korean: K2 PRJEB11591 n = 260, Lee 2024 GSE213647 n = 632, GSE286332 n = 18, Mun 2025 proteome n = 336; Chinese: Mu 2024 HRA004166 n = 214) and **47,587 Asian single cells** (Lu 2023 GSE193581 n = 14,624 malignant cells; Pu 2021 GSE184362 thyrocyte-positive subset n = 32,963) were analysed. The single largest Asian-cohort statistic remains the Mun 2025 proteome **ATC vs PTC dark-matter Fisher OR = 8.54, p = 2.7×10^{-15}** .

A focused single-cell follow-up using Pu et al. 2021 (GSE184362) extracted three primary tumours and three RAI-treated / RAI-refractory metastatic samples (including Patient 11, who developed disseminated subcutaneous and lymph-node disease after three rounds of iodine ablation). All eight panel genes mapped to the 10x feature lists with substantial within-sample heterogeneity. Direction-of-effect at single-sample level was not consistent with the bulk-level refractory < avid pattern (Patient 11 sc met median panel z = +0.45 vs primary-tumour median panel z range 0 to +0.54), most likely reflecting cellular composition differences (thyrocyte-marker-positive cells: Patient 11 sc met n = 5,266 vs primary-tumour samples n = 52–1,808). We disclose this transparently as a single-cell composition caveat; the larger Lu 2023 (n = 14,624 malignant cells) analysis remains the better-powered cellular evidence. See Supp. Fig. S7 and [pu2021_rai_refractory_brief.md](#).

Supplementary Information overview

- **Supplementary Table S1.** Per-sample eight-gene panel scores for TCGA-THCA, GSE151179, GSE299988, GSE112202, Lu 2023 and GSE286332.
- **Supplementary Table S2.** Per-zone HM450 β -values for the 8 panel genes (TCGA-THCA, n = 518).
- **Supplementary Table S3.** Per-zone Cox proportional hazards results (PFI, OS, DSS; TCGA-THCA).
- **Supplementary Table S4.** Mu et al. 2024 four-class driver frequencies and gray-zone composition.
- **Supplementary Table S5.** GSE112202 panel-gene and TDS-16-extra log₂ fold-changes.
- **Supplementary Table S6.** Lee 2024 (GSE213647) per-sample panel z and histology assignments (Korean n = 632).
- **Supplementary Table S7.** Pu 2021 (GSE184362) per-cell panel score across primary tumour and RAI-refractory metastasis samples.
- **Supplementary Figure S1.** Workflow schematic, data sources, label tier mapping.
- **Supplementary Figure S2.** TCGA-THCA per-zone 8-gene RNA z heatmap with full annotations.
- **Supplementary Figure S3.** Eight-gene panel $\subset$ TDS-16 $\subset$ eTDS-64 containment diagram and parsimony AUC comparison.
- **Supplementary Figure S4.** GSE151179 robustness summary.
- **Supplementary Figure S5.** Reviewer-defense scorecard composite.
- **Supplementary Figure S6.** Korean / Asian cohort cross-validation panel (Lee 2024 + K2 + GSE286332 + Mun 2025 + Lu 2023 + Mu 2024 + Pu 2021).
- **Supplementary Figure S7.** Pu 2021 (GSE184362) single-cell follow-up — honest direction-of-effect caveat at single-cell level (under-powered, composition confound).

v2 draft notes: voice-protected sections (Introduction hook, Discussion opening, Limitations subsection, Outlook) have been expanded in this v2 but should be refined by the author before submission. Citations 14, 15, 17, 18, 19 are placeholders pending final journal-format verification.